

Research

Sustainability assessment in higher education institutions: exploring indicators, stakeholder perceptions, and implementation challenges

Najihath Basheer¹ · Vian Ahmed¹ · Zied Bahroun¹ · Chiraz Anane²

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Abstract

Higher Education Institutions (HEIs) play a critical role in addressing global sustainability challenges and advancing the Sustainable Development Goals (SDGs), particularly SDG 4 (Quality Education), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action), through knowledge creation and dissemination. Despite their potential, the integration of sustainability into HEIs is often hindered by fragmented indicators, stakeholder misalignment, and implementation barriers. This study adopts a mixed-methods approach to identify and validate a comprehensive set of Sustainability Indicators (SIs) tailored to the HEI context, with a case study conducted at the American University of Sharjah (AUS). Derived from an extensive literature review, the indicators were validated through semi-structured interviews with key policymakers, revealing gaps in familiarity with sustainability indicators, financial constraints, technological resistance, and decentralized reporting systems. Participants emphasized the importance of environmental initiatives such as energy efficiency and sustainable procurement, while noting the comparatively limited focus on social equity (SDG 10: Reduced Inequalities) and economic dimensions (SDG 8: Decent Work and Economic Growth). Following this, the Relative Importance Index (RII) was employed to analyze stakeholder perceptions through a survey of 81 stakeholders across three key groups (students, educators, and administrative staff) and to rank the indicators by their perceived impact. The findings reveal significant disparities in stakeholder priorities, underscoring the critical need for alignment across governance, environmental, social, economic, and academic pillars. Additionally, challenges such as cost-sustainability trade-offs and fragmented operations highlight the importance of fostering enhanced institutional communication and establishing centralized frameworks to ensure cohesive and effective sustainability efforts. This research offers actionable recommendations to address these barriers, including fostering targeted awareness initiatives, promoting interdisciplinary collaboration, and customizing sustainability frameworks to institutional contexts. By providing a replicable model, the study aligns institutional strategies with stakeholder needs, contributing directly to global sustainability objectives and reinforcing HEIs' role in advancing the UN SDGs.

Keywords Higher education institutions · Sustainability assessment · Sustainability indicators · Stakeholder perceptions · Relative importance index · Sustainability implementation · Sustainable development goals

✉ Zied Bahroun, zbahroun@aus.edu | ¹Department of Industrial Engineering, College of Engineering, American University of Sharjah, Sharjah, United Arab Emirates. ²Department of Foreign Languages, College of Arts, Humanities and Social Sciences, University of Sharjah, Sharjah, United Arab Emirates.



1 Introduction

Sustainability has become a global priority, addressing challenges such as environmental degradation, social inequities, and economic instability. The foundational concept of sustainable development, introduced in the 1987 Brundtland Report, defines it as “development that meets the needs of the present without compromising the ability of future generations to meet their own needs” [1]. This framework underpins global strategies, including the United Nations’ 2015 adoption of the 17 Sustainable Development Goals (SDGs), which address poverty, inequality, climate change, and resource depletion, aiming for achievement by 2030 [2, 3]. Since its inception, sustainable development has evolved, with approximately 300 definitions emerging by 2007, reflecting the complexity of balancing its multidimensional elements [1]. Scholars have noted the vagueness of sustainability definitions [4], while others emphasize adaptability and systemic evolution [5]. Sustainability principles, driven by movements for social justice, conservation, and climate awareness [6], now inform sectors such as agriculture, infrastructure, water management, energy, and higher education institutions.

The 2030 Agenda for Sustainable Development, adopted by the United Nations General Assembly in 2015, provides a roadmap for “peace and prosperity for people and the planet, now and into the future” [7]. This agenda is anchored in the SDGs, which serve as a globally recognized framework for measuring sustainability progress across social, economic, and environmental dimensions. The SDGs not only guide government policies and global sustainability efforts but also provide a benchmark for organizations, institutions, and businesses in aligning their operations with sustainable development. As sustainability gains prominence across different sectors, certification frameworks such as ISO 9001 and ISO 14001 have emerged as tools for measuring and enhancing sustainability performance. These certifications enable institutions to assess their sustainability contributions in a structured and standardized manner, aligning with broader corporate social responsibility (CSR) and SDG objectives [8].

HEIs have emerged as pivotal agents in advancing the sustainability agenda. These institutions are uniquely positioned to foster transformative change by equipping future generations with the knowledge, skills, and values necessary to address sustainability challenges. HEIs serve as incubators for innovation and thought leadership, influencing not only students and faculty but also the broader community. Education for sustainability goes beyond imparting theoretical knowledge; it entails creating awareness about the consequences of unsustainable actions and fostering ethical and responsible behavior. Future leaders, as students today, will shape the trajectory of sustainability in the coming decades through their decisions as professionals, policymakers, and citizens [9, 10]. Thus, integrating sustainability into the educational fabric is not only desirable but imperative.

Despite the growing discourse on sustainability, significant challenges remain in embedding sustainability into higher education. Internationally, systematic reviews and frameworks have emerged to guide the integration of sustainability principles into HEIs. For example, Frizon and Eugénio [9] and De Jesus Lopes et al. [11] emphasize the transformative potential of universities in addressing sustainability through tailored frameworks and performance indicators. However, these efforts often face fragmentation, with indicators and assessment tools dispersed across disciplines and methodologies. Singh et al. [12] and White et al. [13] identify the limitations of current sustainability assessment tools in capturing the multidimensional aspects of sustainability. Sustainability indicators (quantitative or qualitative measures that evaluate and monitor sustainability performance) play a critical role in addressing these challenges. They enable institutions to track progress, inform decision-making, and align practices with sustainability goals [14]. Yet, the diversity of existing frameworks and the lack of standardization continue to hinder effective implementation, underscoring the need for cohesive and adaptable sustainability assessment frameworks tailored to the higher education context.

The implementation of sustainability assessment frameworks in HEIs is further complicated by stakeholder dynamics. Students, educators, and administrative staff often hold divergent views on sustainability priorities and practices, resulting in misaligned sustainability initiatives that hinder cohesive institutional progress. Moreover, external pressures, such as sustainability ranking systems and benchmarking, intensify this complexity, as institutions are increasingly required to demonstrate measurable progress in their sustainability efforts. Addressing these challenges is crucial because HEIs serve as knowledge hubs and play a pivotal role in shaping future leaders, fostering innovation, and influencing societal transformation [9, 11]. Without a unified framework, the potential of HEIs to lead sustainability efforts is significantly undermined. Therefore, there is a critical need for a comprehensive and standardized set of sustainability indicators tailored to the unique context of HEIs. Such a framework must address the multidimensional nature of sustainability while bridging the gaps in stakeholder understanding and engagement [15].

However, while numerous frameworks and assessment tools exist to measure sustainability in HEIs, a significant disconnect remains between sustainability indicators and stakeholder perceptions. Despite efforts to standardize

sustainability assessment, current frameworks often fail to capture the diverse priorities and perspectives of students, educators, and administrative staff. This misalignment can lead to fragmented implementation, inconsistent sustainability policies, and an overall lack of stakeholder engagement. Furthermore, existing sustainability indicators tend to prioritize quantitative metrics, often overlooking the subjective, institution-specific, and perception-based factors that influence sustainability outcomes. Addressing this gap requires a more inclusive and contextually adaptive approach to sustainability assessment.

This study examines sustainability assessment frameworks in HEIs, identifying key SIs and evaluating their relevance through a case study at AUS. Employing a mixed-methods approach, it integrates a literature review, semi-structured interviews with policymakers, and a survey, utilizing the RII to analyze stakeholder perceptions. The research offers actionable recommendations, including targeted awareness initiatives, interdisciplinary collaboration, and customized sustainability frameworks to support HEIs in enhancing sustainability integration and aligning institutional strategies with stakeholder needs. In doing so, this study makes several significant contributions to sustainability in higher education. It develops a comprehensive set of SIs as a practical tool for assessing and improving sustainability performance, identifies key challenges and barriers to implementation, and highlights disparities in stakeholder understanding. By addressing these issues, the study provides a roadmap for HEIs to align their practices with global sustainability goals, fostering a culture of accountability and continuous improvement.

2 Literature review

To ensure a comprehensive review of sustainability assessment in HEIs, a systematic literature retrieval process was employed. The authors conducted a targeted search using the Scopus database between 2018 and 2023, selecting relevant search terms and keywords to identify key studies. The search string used was: “sustainability AND (assessment OR evaluation) AND higher AND education AND (institution OR university) AND (indicator OR framework OR model)”. The selection process considered peer-reviewed journal articles and conference proceedings, prioritizing studies that addressed sustainability indicators, assessment frameworks, and stakeholder engagement in HEIs. Articles were included based on their relevance to sustainability assessment methodologies, institutional implementation challenges, and interdisciplinary approaches. Studies focusing on sustainability in non-educational contexts or lacking empirical validation were excluded.

Sustainability was widely acknowledged as a collective responsibility, with education playing a critical role in advancing sustainable development. Scholars such as Brazier [16] and Chapman et al. [17] emphasized the transformative potential of education in driving global sustainability initiatives. Approaches like Education for Sustainability (EfS) integrated environmental interaction and science-based learning to prepare future leaders for addressing sustainability challenges [18]. HEIs played a pivotal role by promoting ethical and sustainable consciousness while reshaping their missions, curricula, and outreach activities to address contemporary global challenges [11, 19]. These efforts aligned with the environmental, social, and economic pillars of sustainability, emphasizing the interconnected nature of sustainable development.

HEIs increasingly adopted sustainability principles, establishing themselves as hubs of innovation and leadership [9]. Sustainable Development Indicators (SDIs) supported these efforts, serving as metrics for assessing change and performance [20–22]. By linking SDIs to specific goals, institutions enhanced their impact, drawing on historical practices, regulatory frameworks, and expert insights [23–25]. During the early adoption phases, universities tailored sustainability tools to their unique contexts, ensuring relevance and practicality [26]. Comprehensive reviews, such as those by [25], evaluated Sustainability Assessment Tools (SATs) used in HEIs, identifying critical indicators and challenges. Maragakkis and Dobelsteen [27] proposed refinements to these tools to improve standardization and application. Despite these efforts, fostering cohesive sustainability practices remained challenging, underscoring the need for further refinement [28, 29].

Research into the implementation of sustainability in HEIs demonstrated notable progress in aligning institutional practices with SDGs. These efforts spanned the three pillars of sustainability, highlighting their interconnectedness. Griebeler et al. [30] proposed a framework of indicators to measure HEIs' contributions to SDGs, offering a structured approach to assessment. Zahid et al. [31] introduced the Sustainability Integration and Implementation Index to evaluate institutional practices, while Bautista-Puig and Sanz-Casado [32] explored sustainability integration within Spanish universities, noting both synergies and disparities between public and private institutions. Gutiérrez-Mijares et al. [33] provided a global review of methodologies for assessing sustainability, identifying recurring themes and approaches.

Simina [34] examined how sustainability-focused research enhanced university competitiveness. Building on these insights, Caeiro and Azeiteiro [35] assessed the effectiveness of sustainability strategies implemented across diverse institutional contexts, further supporting the integration of the three pillars into institutional practices.

Emerging technologies are transforming sustainability efforts in HEIs, enabling smarter data collection, monitoring, and strategic decision-making. Digital transformation plays a key role in streamlining reporting, optimizing resource use, and enhancing campus sustainability initiatives. Aziz et al. [36] highlight how digital mapping technologies support sustainable infrastructure, offering valuable insights for HEIs in sustainability assessments and planning. Similarly, Almuaythir et al. [37] emphasize how robotics and automation can improve efficiency, reduce resource consumption, and enhance sustainability education, aligning with HEIs' efforts to integrate technology into both curricula and operations. Shivendra et al. [38] explore 3D concrete printing's sustainability potential while highlighting financial and regulatory barriers, similar to those faced by HEIs in adopting innovative sustainability solutions. By leveraging these technologies, HEIs can enhance academic and operational sustainability, ensuring data-driven and adaptive decision-making.

Beyond technological advancements, sustainability challenges require a multidisciplinary approach. Addressing them effectively demands integrating expertise from engineering, business, and environmental sciences to develop holistic solutions. Martins et al. [39] advocate for interdisciplinary learning models that foster critical thinking and adaptability, helping HEIs move beyond isolated sustainability initiatives toward comprehensive strategies addressing environmental, social, and economic sustainability. Similarly, Petrović et al. [40] highlight how embedding circular economy principles into curricula fosters long-term economic resilience and sustainability. By integrating emerging technologies and interdisciplinary approaches, HEIs can bridge the gap between sustainability education and real-world application. As hubs for innovation, they play a critical role in ensuring sustainability assessments, institutional practices, and academic programs remain dynamic, data-driven, and aligned with global sustainability goals.

Despite advancements, significant barriers impeded the effective implementation of sustainability initiatives across the environmental, social, and economic pillars. Institutional resistance, resource constraints, and insufficient leadership hindered progress. Many institutions struggled to embed sustainability cohesively due to a lack of strategic alignment. Veidemane [41] highlighted the complexity of developing universally acceptable sustainability indicators, while Alsharif et al. [42] identified cultural and organizational challenges specific to Saudi Arabia. Zainordin and Ismail [43] critiqued existing tools for their inability to address the full scope of sustainability practices, emphasizing the need for comprehensive frameworks to overcome institutional barriers and foster interdisciplinary collaboration.

The environmental pillar emphasized ecological stewardship through green initiatives, carbon footprint reduction, and sustainable transportation. Ali and Agbozo [45] examined the positive impact of campus sustainability initiatives on environmental quality, while Obrecht et al. [46] proposed frameworks for embedding sustainability into educational programs. Menon and Suresh [47] developed conceptual models using fuzzy logic to refine environmental practices in Indian universities. Sanchez et al. [48] employed multi-criteria methods to prioritize campus sustainability indicators, offering data-driven approaches to enhance institutional decision-making. Innovative solutions such as smart campus systems and real-time monitoring dashboards showed promise, as demonstrated by Dalla Gasperina et al. [49], though gaps persisted in standardizing criteria and integrating digital sustainability measures, including those for reducing the carbon footprint of digital activities [50].

The social pillar addressed equity, leadership, and community engagement, focusing on fostering inclusive governance and collaboration. Pizzutilo and Venezia [10] proposed frameworks for integrating social responsibility into HEIs, while Pedro, Leitão, and Alves [51] explored how intellectual capital impacted stakeholder quality of life. Research into Green Human Resource Management (GHRM), as demonstrated by Asfahani [52], highlighted its role in enhancing environmental performance and competitive advantage. Lima et al. [53] and Niedlich et al. [54] emphasized the importance of governance in achieving social sustainability, while Fahim et al. [55] identified organizational change as critical for transforming HEIs into socially responsible institutions. These findings underscored the importance of exploring Environmental, Social, and Governance (ESG) practices to promote long-term social sustainability.

The economic pillar focused on financial stability, resource allocation, and sustainable investments. Studies by Cernostana [56] and Fedulov et al. [57] emphasized the importance of quantitative assessments for evaluating financial sustainability in private institutions, while Abad-Segura and González-Zamar [58] examined how HEIs contributed to economic sustainability within the SDG framework. The absence of standardized methodologies and benchmarks for comparing financial sustainability across diverse HEIs posed a significant challenge. Addressing these gaps required integrating financial metrics with broader sustainability indicators, fostering a holistic approach to institutional planning and management. Innovative methodologies such as cognitive mapping offered potential for enhancing assessment accuracy and relevance.

The current literature on sustainability assessments in HEIs reflects significant progress in integrating sustainability dimensions across environmental, educational, social, and economic domains. While environmental sustainability and its associated assessment tools have received substantial attention, research on the social and economic dimensions remains comparatively limited. Existing sustainability assessment tools, such as STARS, the UI Green Metric, and Times Higher Education Ranking systems, dominate evaluations of sustainability in HEIs. However, these tools often fall short in comprehensiveness, particularly in addressing strategic and policy-related elements and adapting to nuanced regional and institutional needs.

The reviewed studies emphasize the widespread adoption of frameworks and indicators for sustainability assessment. Yet, the implementation of these frameworks is frequently hampered by challenges such as institutional inertia, limited stakeholder awareness, and inconsistencies in sustainability competencies. The complexity and resource-intensive nature of sustainability assessments further exacerbate these challenges. Additionally, significant gaps persist in achieving consensus on comprehensive sustainability indicators and tailored assessment tools that can adapt to diverse contexts and provide insights into the long-term impacts of sustainability initiatives.

To address the gaps identified in the literature, this research explored existing studies to compile a comprehensive set of sustainability indicators relevant to HEIs. Derived from a detailed review of scholarly works, these indicators provide the foundation for a mixed-methods approach that considers the multidimensional nature of sustainability. Grounded in the three pillars of sustainability (environmental, economic, and social), these indicators ensure a holistic evaluation of sustainability practices and align with institutional goals for achieving sustainable development.

The indicators are categorized into five core dimensions, presented in Table 1, which collectively address the three sustainability pillars. The first dimension, Governance and Operations, aligns primarily with the social and economic pillars. It addresses institutional frameworks, strategies, and policies that drive sustainability initiatives, including sustainability integration into governance structures, strategic plans, and stakeholder engagement. These elements ensure sustainability is embedded in decision-making processes and operational frameworks, reflecting the institution's commitment to long-term viability and social responsibility.

The second dimension, Environmental Impact and Resource Management, directly corresponds to the environmental pillar. This dimension emphasizes minimizing the ecological footprint of HEIs through biodiversity conservation, energy efficiency, waste management, and pollution reduction. Innovative practices, such as renewable energy use and water conservation, also fall within this dimension, showcasing institutional efforts to prioritize environmental stewardship.

Economic Viability and Affordability represents the economic pillar of sustainability, focusing on financial stability, resource allocation, and equitable distribution of resources. Indicators such as financial assistance programs, ethical costing systems, and affordability policies for low-income students ensure sustainability initiatives are economically feasible and accessible. This dimension highlights the importance of integrating financial considerations into sustainability practices to promote inclusivity and fairness.

The fourth dimension, Academic Integration, bridges the three pillars by fostering sustainability education and research. It emphasizes the integration of sustainability principles into curricula, experiential learning, and research activities. Indicators in this category include sustainability literacy assessments, graduate programs in sustainability, and interdisciplinary collaboration. This dimension ensures HEIs equip students and faculty with the knowledge and skills necessary to address sustainability challenges comprehensively.

Finally, Social Responsibility and Equity focuses on the social pillar, addressing issues such as equity, diversity, community engagement, and well-being. Indicators include gender equality initiatives, healthy and ergonomic work environments, and collaborations with societal stakeholders. This dimension emphasizes fostering inclusive and equitable practices within and beyond the institution, promoting a culture of social sustainability.

By linking these five dimensions to the three pillars of sustainability, this framework provides a structured approach for evaluating sustainability practices in HEIs. The integration of expert validation through interviews and the classification of stakeholder perceptions using the RII ensures the framework is adaptable to diverse institutional contexts and aligned with global sustainability objectives.

Figure 1 was developed based on an extensive review of existing sustainability assessment frameworks and indicators used in HEIs. It presents a synthesized framework that categorizes key sustainability indicators into five core dimensions: Governance and Operations, Environmental Impact and Resource Management, Economic Viability and Affordability, Academic Integration, and Social Responsibility and Equity. This framework provides a structured approach to evaluating and implementing sustainability practices across diverse institutional domains.

The indicators will first be proposed to experts for validation through semi-structured interviews, ensuring their relevance, applicability, and comprehensiveness. Expert feedback will refine the indicators to address specific institutional

Table 1 Categorization of Sustainability Indicators in Higher Education Institutions

Sustainability dimensions	Key indicators
Governance and operations [10, 30, 31, 42, 44, 59–61, 64, 66]	SD budget and in-kind support, Policy for implementing SD, Strategic plan for implementing sustainability, SD office/department, SD included in the vision and mission, goals and objectives, Sustainability updates on the website, Stakeholder identification and engagement, Sustainable procurement policies, Sustainable transportation, Efficient information infrastructure, Assessment of SD issues, Communication of SD activities, Sustainability reports, National and international SD universities rankings'
Environmental Impact and resource management [10, 30, 31, 42, 44, 60–62, 64–66]	Conservation of biodiversity in and around campus, Landscape management, Plans to improve energy efficiency, Energy-efficient equipment, Renewable energy availability & usage, Plans and efforts to reduce GHG emissions, Plans to improve the reduction and management of waste, Hazardous waste management, Waste bins to separate and recycle waste, Pollution reduction and recycling activities, Rainwater management and conservation, Plans for water and wastewater management, Green Initiatives and sustainable practices, Green buildings initiative, Efforts to reduce food waste, Policies and activities to reduce paper consumption, Hiring skilled environmental professionals, Formation of environmental research centers and committees, Smart campus system implementation
Economic Viability and affordability [10, 30, 31, 42, 44, 61, 62, 64, 66]	Policies for affordability for low-income students, financial assistance from government, Financial/non-financial support and incentive programs to students, Ethical and life cycle costing systems, Funding/Grants & contracts for SD research
Academic Integration [10, 30, 31, 42, 60–63, 66]	Integrated sustainability courses throughout all curricula (mandatory and optional), Graduate program on sustainability, Availability of courses for e-learning, Continuing education to the public on SD, Courses to educate the educators on SD teaching, Sustainability literacy assessment for students, Experiential learning, Soft skills training activities, Use of research generated in SD teaching, Promotion of holistic thinking in research, Staff expertise on sustainability (attend an event or course), SD institute research center, Publications on SD, Patents in the field of SD, Creation of SD new knowledge and technologies, Open access to research, Invited and permanent Staff dedicated to SD, Courses for vocational/academic training for employees, Teacher's performance, expertise, punctuality, and respectfulness, Infrastructure and Campus Environment
Social Responsibility [10, 30, 31, 42, 44, 61, 62, 64, 66]	Healthy lifestyle promotion and health services, Healthy and ergonomic work environments, Diversity and equity, Gender equality, Exchange programs in SD, Joint SD degrees with other universities, Joint SD research with other universities, SD partnerships with other society stakeholders, Part of interdisciplinary SD expert networks, Collaboration with NGOs and consultants to anticipate future needs and demands, Academic staff involved in voluntary advisory activities in SD, SD events that are open to the public/community, Knowledge sharing using media (email, bulletin, conference), Focus on social, environmental, and economic sustainability, Sustainable practices for students, Student participation in SD activities, Employees and other stakeholders act sustainably, SD efforts are visible throughout the campus, Staff and researchers are aware of SD, Training, counselling, prevention, and risk-control programs, Cultural or scientific initiatives outside the community, SD working group with members from different departments

needs. Subsequently, stakeholder perceptions of the validated indicators will be assessed through surveys. Using the RII, the study will classify and rank these indicators to identify priority areas across different stakeholder groups, including students, educators, and administrative staff. This process not only highlights the relative importance of each indicator but also enables a comparative analysis of stakeholder perspectives, ensuring adaptability while aligning institutional practices with global sustainability objectives.

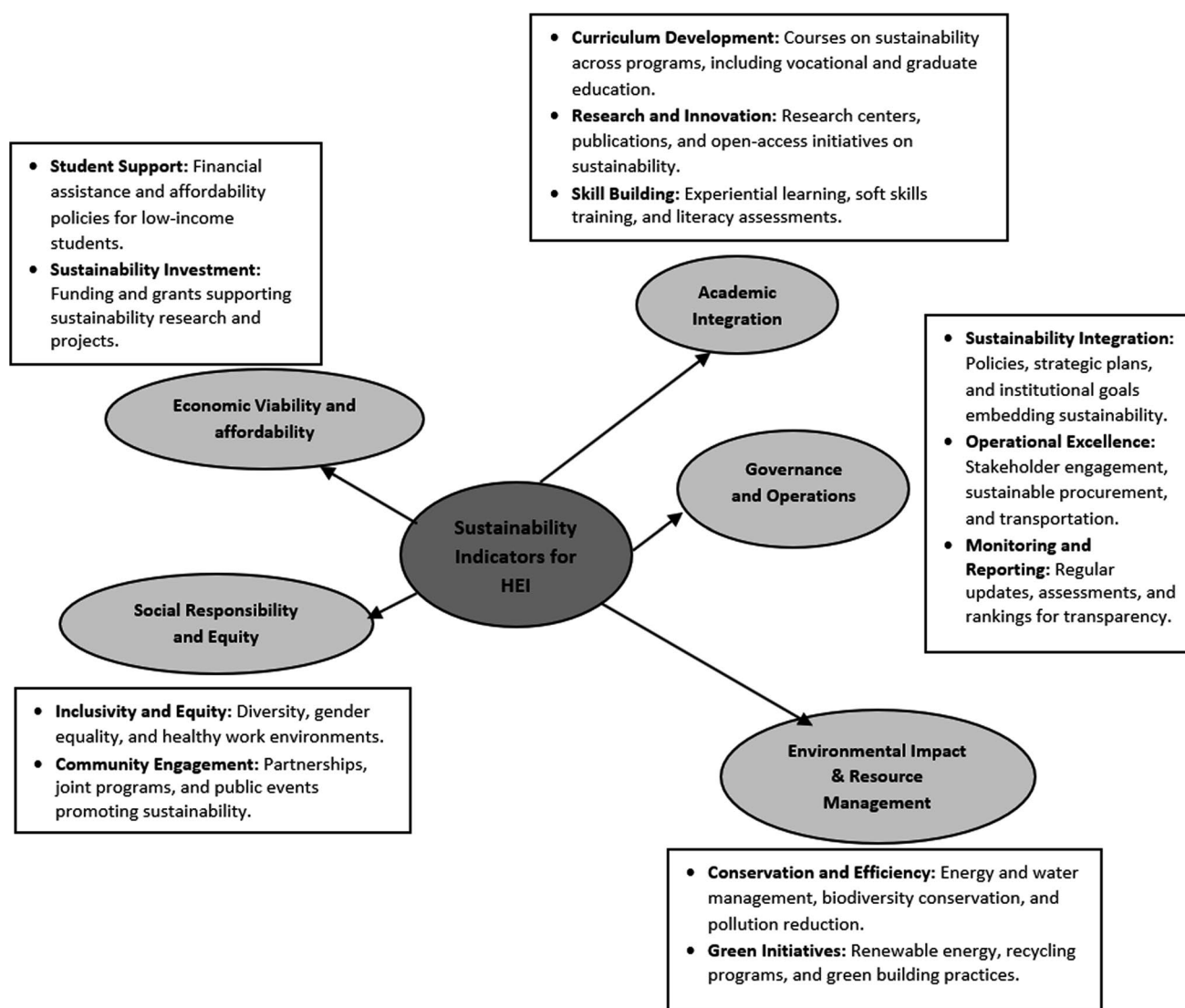


Fig. 1 Sustainability Indicators Framework for Higher Education Institutions

3 Research methodology

This study adopts a mixed-methods approach, integrating qualitative and quantitative techniques to validate and refine sustainability indicators for HEIs. The methodology combines a comprehensive literature review, semi-structured expert interviews, and stakeholder surveys, aiming to identify and analyze disparities in the perception of sustainability indicators among students, educators, and administrative staff. The findings are contextualized using a case study of a higher education institution. An abductive reasoning framework is employed, combining inductive methods to identify sustainability indicators and deductive methods to validate and rank them. The study's exploratory phase identifies key sustainability indicators and challenges through an extensive literature review and experts' interviews, while the descriptive phase quantifies stakeholder perceptions via surveys and RII analysis. The case study of the American University of Sharjah (AUS) provides contextual grounding for the research findings.

The study complies with ethical guidelines approved by the Institutional Review Board (IRB) at the chosen HEI. Participants are required to provide informed consent before participation. Confidentiality and anonymity are maintained throughout the research process. Secondary data sources are properly cited to ensure academic integrity.

The main steps of the research methodology are outlined below:

– Data collection

- Semi-structured expert interviews:

Experts validate the indicators derived from the literature and identify additional challenges in implementing sustainability practices. The semi-structured format provides flexibility while maintaining consistency. Insights from these interviews refine the list of indicators and address the challenges policymakers face.

- Stakeholder Surveys:

Surveys are distributed among students, educators, and Administrative/Governance/Operations (AGO) staff to evaluate the perceived importance of sustainability indicators. To ensure the validity and reliability of the survey, the indicators were first reviewed and validated by experts during the semi-structured interviews, ensuring their relevance to HEIs. Reliability was assessed through a structured questionnaire using a 5-point Likert scale, with the RII applied to maintain consistency in stakeholder responses. To mitigate respondent bias, the survey was anonymous, included a balanced representation of students, educators, and administrative staff, and provided clear definitions of sustainability indicators to ensure informed responses. Additionally, distributing the survey online allowed participants to respond independently, reducing potential social desirability or time-related biases. Convenience sampling is used for accessibility, while ensuring diverse representation.

– Data Analysis

The RII method ranks sustainability indicators based on stakeholder responses, calculated as:

$$RII = \sum w / (A * N) \quad (1)$$

where w is the weight assigned by respondents, A is the maximum weight (5), and N is the total number of respondents. The resulting RII values are categorized into five levels of importance, ranging from high to low [67, 68].

The RII approach was selected for ranking sustainability indicators due to its effectiveness in quantifying stakeholder perceptions and prioritizing multiple factors based on their perceived importance. RII has been widely applied in multi-criteria decision-making studies, particularly in fields such as construction management, education, and sustainability research [67, 68], where stakeholder input is diverse and subjective. Its suitability for sustainability assessment in HEIs stems from its ability to compare the relative significance of various sustainability indicators across different stakeholder groups (students, educators, and AGO staff), providing a structured method for evaluating institutional priorities. Unlike simple mean ranking methods, RII ensures normalization of responses, mitigating scale biases and facilitating a consistent comparison. This allows institutions to identify key sustainability areas requiring intervention and ensures that sustainability strategies align with stakeholder priorities.

4 Results and analysis of sustainability practices in HEIs

This section presents the findings on sustainability practices within the HEI, focusing on key sustainability indicators across governance, environmental, social, economic, and academic dimensions. The analysis begins with insights from policymakers, who provided perspectives on institutional sustainability initiatives, the relevance of identified indicators, and the challenges encountered in implementation. This is followed by an examination of survey responses, which highlight stakeholder perceptions and the prioritization of sustainability indicators. The results offer a comprehensive understanding of sustainability efforts within the institution, integrating both policy-level insights and stakeholder viewpoints to inform actionable recommendations in later sections.

4.1 Semi structure interviews

Policymakers provided key insights into the institution's sustainability efforts, discussing their awareness of global sustainability rankings and evaluating the relevance of identified sustainability indicators. They highlighted ongoing and planned sustainability initiatives within their departments, detailing specific projects aimed at advancing sustainability objectives. Additionally, they identified significant challenges hindering implementation, including financial constraints, technological limitations, and institutional resistance.

Eight policymakers, each with a minimum of five years of experience, participated in these interviews. They represented diverse departments, including Finance, Supply Chain, Campus Development, Facilities, Student Affairs, Safety and Crisis Management, Research Services, and Curriculum Development. The insights from these interviews are analyzed in detail in the following subsections.

4.1.1 Awareness of institutional sustainability efforts

The interviews revealed that only three out of eight participants were fully aware of the institution's participation in global sustainability rankings, such as the Times Higher Education and AASHE Sustainability Ratings. For instance, one participant mentioned, *"Yes, I am a sustainability researcher and know about the rankings and AUS's participation"*, while another stated, *"Yes, I am aware of the ranking systems and the HEI is participating in them"*. However, other participants exhibited limited or no awareness of these efforts, as highlighted by one interviewee's response: *"I have heard about rankings, but not aware of the HEI is participating in these ranking systems"*.

This gap highlights a lack of consistent communication across departments, underlining the need for enhanced strategies to ensure all stakeholders understand the significance of sustainability rankings for institutional reputation and progress. Participants who were aware of the rankings emphasized their importance in positioning the institution as a leader in sustainability efforts. These insights demonstrate the necessity of embedding sustainability knowledge into regular departmental communications and training programs, fostering a culture of awareness and alignment with institutional goals.

4.1.2 Awareness and validation of sustainability indicators

The interviews revealed varying levels of familiarity with sustainability indicators among participants. Some interviewees demonstrated a clear understanding, with one noting, *"Yes, I know about the sustainability indicators"*. Another participant highlighted ongoing efforts in this area, stating, *"Yes, we have ongoing research involving sustainability indicators"*. Others exhibited partial awareness, as reflected in the comment, *"Yes, I have heard about the indicators but have only partial knowledge"*. Conversely, several participants admitted to having no prior exposure to the concept, stating, *"No, this is my first time,"* or *"No, it's a completely new concept for me."*

Despite these disparities, all participants validated the indicators' relevance and recognized their importance in guiding sustainability efforts. The findings emphasize the need for institution-wide efforts to enhance understanding and communication regarding sustainability indicators, ensuring stakeholders are equipped to apply these measures effectively in their respective roles.

The variability in familiarity highlights a gap in institutional communication and education regarding sustainability indicators. Addressing this gap is critical to ensuring that all stakeholders are equipped to integrate these indicators effectively into their respective practices.

The findings point to a pressing need for institution-wide initiatives to enhance understanding and engagement with sustainability indicators. Targeted workshops, training sessions, and consistent communication efforts could play a pivotal role in building this shared knowledge base. By fostering a unified understanding of sustainability indicators, institutions can better align stakeholder priorities, ensure more consistent application of sustainability measures, and create a cohesive framework for assessment and implementation.

4.1.3 Sustainability initiatives

The interviews revealed a diverse array of sustainability initiatives undertaken by various departments, with a primary emphasis on environmental sustainability. Key actions included digitalization efforts to reduce paper usage, implementing energy-efficient practices to minimize power consumption, eliminating single-use products, and adopting sustainable procurement policies. Several departments highlighted their commitment to these practices by integrating technology into their operations. As one interviewee noted, *"We are focusing on digitalization and eliminating single-use products while ensuring that our suppliers follow sustainable practices"*. Another respondent emphasized their department's transition to digital platforms, stating, *"We have developed software for approvals and queries instead of paper requests, which has significantly reduced our reliance on printed documents"*.

Specific environmental initiatives were also reported, including the installation of solar panels for water heating, tinting windows to regulate indoor temperatures, and encouraging carpooling to reduce energy consumption. As one interviewee described, *"We have installed solar panels in dormitories to support water heating and are also monitoring power usage to enhance energy efficiency"*. These efforts reflect a proactive approach to reducing the environmental footprint across institutional operations.

Community and student engagement also emerged as notable aspects of sustainability efforts. The Student Affairs department actively supports student-led sustainability initiatives, recognizing sustainable practices at events and promoting reduced plastic use and food waste. One interviewee highlighted these efforts, stating, *"We monitor sustainability during events, construct reusable tents, and support student sustainability clubs"*.

However, broader dimensions of sustainability, particularly those related to social equity and economic viability, received comparatively less attention. While some departments outlined plans to integrate sustainability into curricula or research activities, these efforts were less pronounced than environmental initiatives. One respondent acknowledged this gap, stating, *"New courses on sustainability are being introduced in all disciplines, but more work is needed to embed sustainability holistically across research and teaching"*.

These insights highlight the fragmented nature of sustainability initiatives across departments, with significant variation in priorities and execution. To address these gaps, institutions could implement a more integrated strategy that emphasizes the multidimensional nature of sustainability, ensuring balanced efforts across environmental, social, and economic domains.

4.1.4 Challenges in implementing sustainability initiatives

The interviews revealed several recurring challenges faced by departments in implementing sustainability initiatives. A prominent issue was financial constraints, with participants noting difficulties in balancing sustainability goals with budgetary limitations. Many departments found it challenging to justify long-term investments in sustainability initiatives without immediate returns. One participant described this dilemma as a *"cost-sustainability trade-off"* adding that it is a *"time-consuming process with difficulty in quantifying the initiatives"*. Another interviewee highlighted similar concerns, stating, *"High costs and managing a large campus area make sustainability efforts even more challenging"*.

Another major challenge was the lack of awareness and education about sustainability practices. Participants emphasized the need for ongoing training and capacity-building programs to foster a culture of sustainability. One interviewee noted, *"Creating awareness about sustainability remains one of the biggest hurdles"*. This gap in awareness highlights the importance of equipping staff and students with the knowledge and skills required to actively engage in sustainability efforts.

Technological adaptation also emerged as a significant barrier. Departments reported struggles in keeping up with rapidly evolving technologies and overcoming resistance to change. One respondent explained, *"Keeping up with technology and overcoming resistance to change has been a key challenge"*, while another interviewee emphasized the difficulty of adapting to technological advancements, stating, *"Catching up with technology remains a major issue"*. These responses indicate that while digital tools and innovations are critical to sustainability efforts, a lack of familiarity and acceptance slows their adoption.

Coordination and reporting issues further complicated sustainability implementation. The absence of centralized systems for sustainability reporting led to fragmented efforts across departments, making it difficult to track progress and align initiatives institution-wide. This lack of cohesion underscores the need for improved systems to streamline reporting and foster collaboration among departments.

Interestingly, while most participants acknowledged these challenges, two departments reported facing no significant hurdles in implementing sustainability initiatives. One interviewee simply stated, "*No major challenges yet*", suggesting that variations in departmental engagement, priorities, or resources could influence the perceived difficulties of implementing sustainability efforts.

These challenges underscore the importance of institutional strategies to systematically address these barriers. Allocating dedicated budgets for sustainability, implementing training programs, and developing centralized reporting systems could significantly enhance the effectiveness of sustainability initiatives.

The validation of sustainability indicators by the participants reinforces their relevance and applicability in decision-making and operational activities. The insights gathered from the interviews complement the findings from the literature review, providing a comprehensive foundation for the next phase of this research. These findings, along with the indicators identified through the literature review, informed the development of the survey instrument used in the subsequent section. This approach ensures that the survey reflects both theoretical frameworks and practical perspectives, capturing a holistic view of sustainability priorities and challenges within the studied HEI.

4.2 Online questionnaire surveys

Online surveys were conducted across the institution to evaluate how different stakeholders perceive and rate various sustainability indicators in HEIs based on their understanding. The stakeholders were categorized into three groups: students, educators, and AOG staff. The questionnaire, distributed online via Google Forms, comprised 20 questions covering participant consent, demographic profiles, and their understanding of sustainability indicators for sustainability assessments in HEIs. A total of 81 responses were collected and are systematically analyzed in the following section.

4.2.1 Demographic analysis

The demographic analysis, as detailed in Fig. 2, reveals the following insights:

- The largest age group among respondents is 35–50 years, comprising 30.9% of the total.
- Females represent 55.6% of respondents, indicating a slight predominance over males.

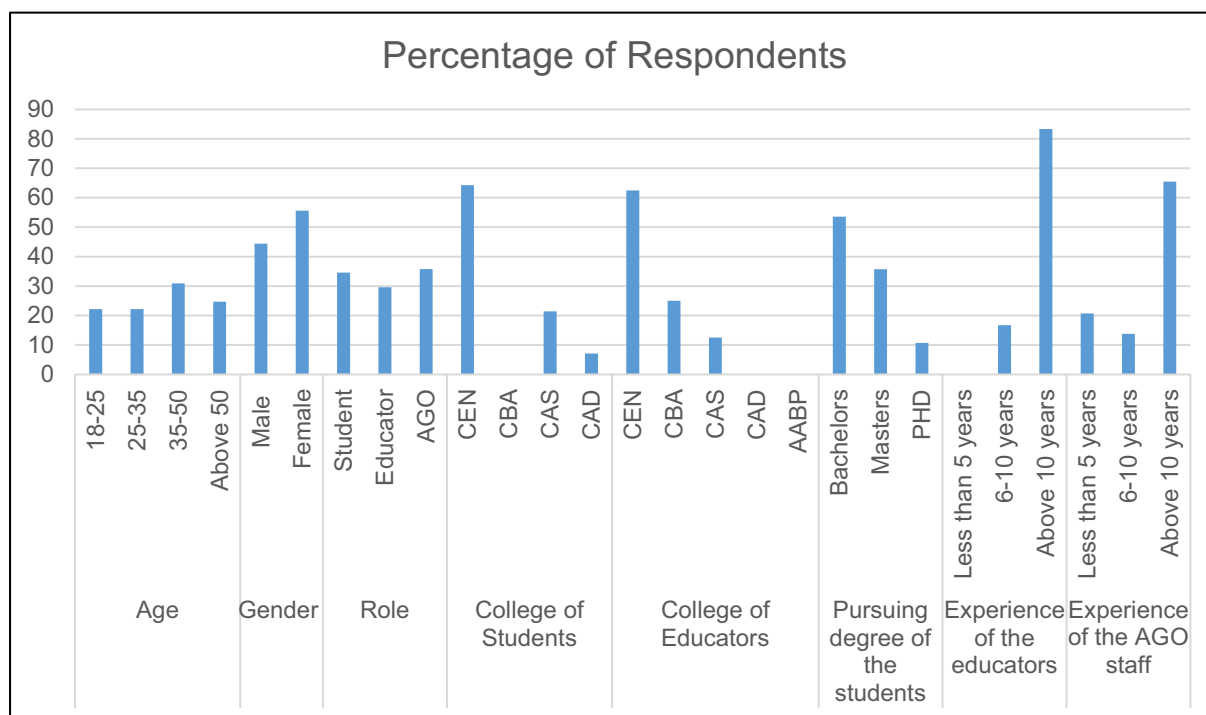


Fig. 2 Demographic Profile of Survey Respondents

- The respondents are categorized into students (34.6%), educators (29.6%), and Administration, Governance, and Operations (AGO) staff (35.8%).
- Among students, a significant 64.3% are from the College of Engineering (CEN).
- For educators, 62.5% are associated with the College of Engineering (CEN).
- A majority of students, 53.6%, are pursuing bachelor's degrees.
- Notably, 83.3% of educators have over 10 years of experience.
- Among AGO staff, 65.5% have more than 10 years of professional experience.

These findings highlight a diverse respondent profile with substantial representation from the College of Engineering and a workforce possessing significant experience.

4.2.2 Assessment of stakeholder understanding of sustainability indicators

This section of the survey analyses stakeholders' understanding of the identified sustainability indicators using a 5-point Likert scale. Respondents (students, educators, and AGO staff) rated indicators across five distinct categories, and the RII analysis was employed to determine which indicators within each category were rated highest by the HEI stakeholders. The results of the RII analysis for each category are detailed in the subsequent sections, offering insights into the indicators deemed.

4.2.2.1 Category 1: governance and operations This section evaluates 14 sustainability indicators under the category of governance and operations. Stakeholders (students, educators, and AGO staff) rated these indicators using a 5-point Likert scale, and the RII was calculated for each group separately. The results are summarized in Table 2, providing a clear comparison of stakeholder perspectives. The highest-rated indicator for each stakeholder group is highlighted in bold red in the table. Figure 3 visually represents the variation in RII values for the sustainability indicators across the three groups.

Table 2 indicates that the highest-rated indicator among students and educators is "Strategic plan for implementing sustainability", with RII values of 0.764 and 0.767, respectively. For AGO staff, the top-ranked indicator is "SD included in the vision, mission, goals, and objectives", with an RII value of 0.897, highlighting its perceived criticality. AGO staff consistently rated most indicators higher than students and educators, reflecting their stronger emphasis on governance and operational factors.

From Fig. 3, it is evident that AGO staff consistently rated sustainability indicators higher across the board, particularly in areas like "Sustainability Development (SD) budget and in-kind support", "Policy for implementing SD", and "SD included in the vision, mission, goals, and objectives", indicating their prioritization of foundational governance structures. Educators showed a balanced approach, giving medium to high importance to most indicators, including "Sustainability updates on the website" and "Efficient information infrastructure", which reflect their role in communication and

Table 2 RII for sustainability indicators by the stakeholders- Category 1

No	Indicators: Category 1	Students	Educators	AGO Staff
1	Sustainability Development (SD) budget and in-kind support	0.700	0.742	0.828
2	Policy for implementing SD	0.757	0.725	0.841
3	Strategic plan for implementing sustainability	0.764	0.767	0.862
4	SD office/department	0.721	0.742	0.828
5	SD included in the vision and mission, goals, and objectives	0.764	0.775	0.897
6	Sustainability updates on the website	0.643	0.758	0.821
7	Stakeholder identification and engagement	0.729	0.683	0.807
8	Sustainable procurement policies	0.700	0.700	0.814
9	Sustainable transportation	0.721	0.717	0.807
10	Efficient information infrastructure	0.650	0.692	0.814
11	Assessment of SD issues	0.714	0.725	0.772
12	Communication of SD activities	0.714	0.733	0.800
13	Sustainability reports	0.679	0.683	0.800
14	National and international SD universities rankings	0.679	0.708	0.800

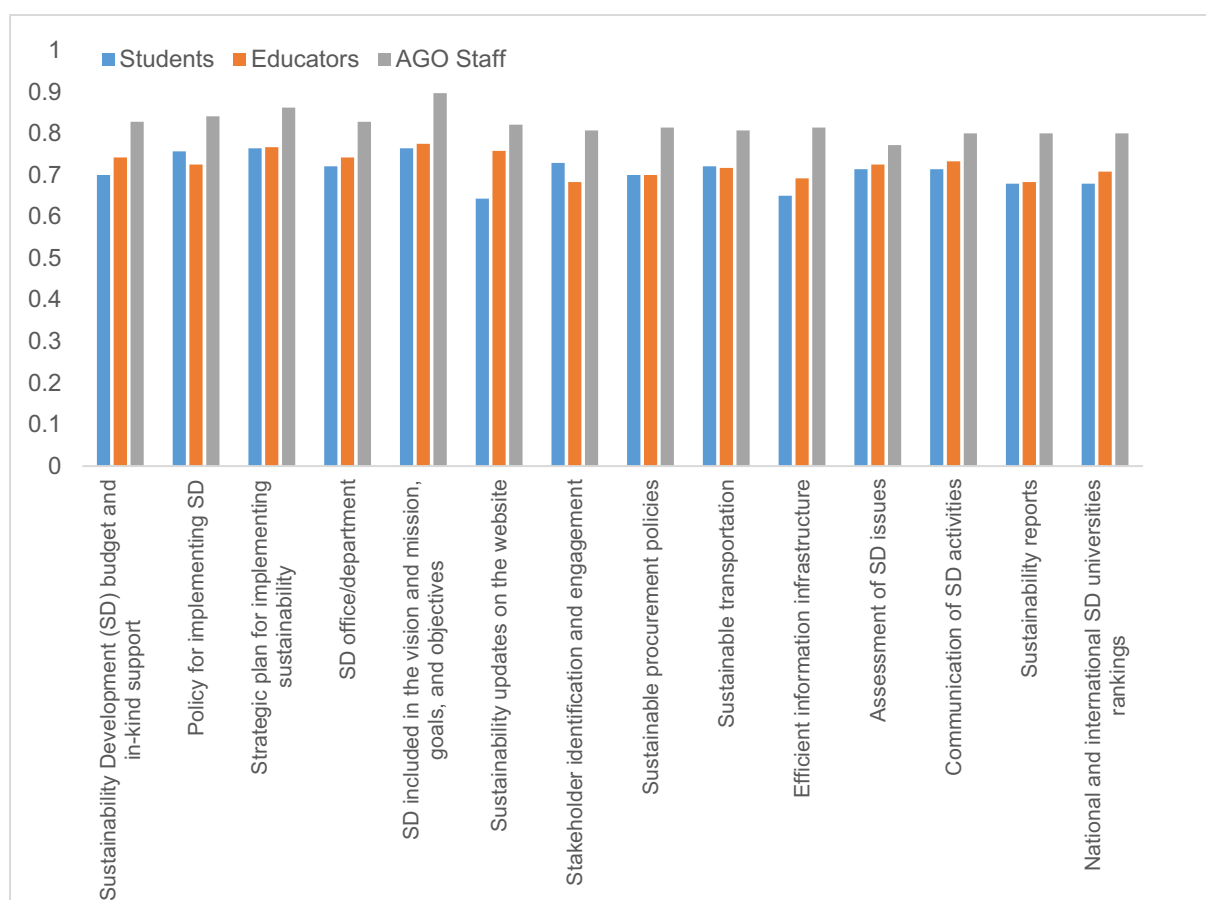


Fig. 3 Chart showing RII for category 1 sustainability indicators

resource management. Students, on the other hand, rated several indicators lower than the other groups, with the lowest ratings for "Sustainability updates on the website" (RII = 0.643) and "Efficient information infrastructure" (RII = 0.650), suggesting a gap in their engagement or awareness regarding these aspects.

While "Strategic plan for implementing sustainability" emerges as a common priority among students and educators, AGO staff place greater emphasis on aligning sustainability with the institution's vision, mission, goals, and objectives. This difference in perspectives underscores the varying roles and responsibilities of each stakeholder group in sustainability governance. Importantly, the data highlights a disconnect between policymakers' strategic priorities and the awareness levels of students, emphasizing the need for better communication and integration of sustainability initiatives across all stakeholder groups.

Figure 3 below illustrates the variation in RII for each sustainability indicator of category 1.

4.2.2.2 Category 2: environmental impact and resource management This section analyses 19 sustainability indicators related to environmental impact and resource management. Each group's RII was determined, and the findings, summarized in Table 3, highlight comparative stakeholder priorities. In the table, the most significant indicator for each stakeholder group is marked in bold red. Additionally, Fig. 4 visually presents the variations in RII values among the three groups.

The results show distinct priorities among the stakeholder groups. Table 3 indicates that students rated "Plans for water and wastewater management" the highest, with an RII value of 0.771, reflecting their focus on resource conservation. Educators assigned the highest importance to "Waste bins to separate and recycle waste", with an RII value of 0.825, emphasizing their commitment to effective waste management practices. AGO staff, on the other hand, prioritized "Plans to improve energy efficiency", which achieved the highest RII value of 0.890, highlighting their emphasis on planning and energy optimization initiatives.

Table 3 RII for various sustainability indicators by stakeholders- Category 2

No	Indicators: Category 2	Students	Educators	AGO Staff
1	Conservation of biodiversity in and around campus	0.686	0.742	0.821
2	Landscape management	0.736	0.800	0.814
3	Plans to improve energy efficiency	0.743	0.800	0.890
4	Energy-efficient equipment	0.707	0.775	0.848
5	Renewable energy availability & usage	0.736	0.725	0.814
6	Plans and efforts to reduce GHG emissions	0.736	0.717	0.821
7	Plans to improve the reduction and management of waste	0.764	0.767	0.848
8	Hazardous waste management	0.743	0.792	0.862
9	Waste bins to separate and recycle waste	0.757	0.825	0.883
10	Pollution reduction and recycling activities	0.743	0.775	0.821
11	Rainwater management and conservation	0.721	0.642	0.697
12	Plans for water and wastewater management	0.771	0.683	0.766
13	Green Initiatives and Sustainable Practices:	0.750	0.783	0.800
14	Green buildings initiative	0.729	0.700	0.821
15	Efforts to reduce food waste	0.693	0.717	0.745
16	Policies and activities to reduce paper consumption	0.686	0.750	0.766
17	Hiring skilled environmental professionals	0.700	0.667	0.745
18	Formation of environmental research centers and committees	0.750	0.750	0.772
19	Smart campus system implementation	0.750	0.700	0.752

The visualization in Fig. 4 reveals that AGO staff consistently assigned higher RII values to indicators compared to students and educators, particularly for indicators such as “Plans to improve energy efficiency”, “Waste bins to separate and recycle waste”, and “Hazardous waste management”. This reflects their role as policymakers and their focus on strategic initiatives. Educators showed balanced prioritization across most indicators, with a notable emphasis on waste management and green initiatives, such as “Green Initiatives and Sustainable Practices” and “Waste bins to separate and recycle waste”. Students demonstrated relatively lower prioritization, with medium to high importance assigned to indicators like “Plans for water and wastewater management” and “Waste bins to separate and recycle waste”.

The analysis highlights a key divergence in perspectives: AGO staff, as policymakers, emphasize planning and long-term energy efficiency, while educators focus on immediate operational actions such as waste management. Students, though engaged, tend to prioritize practical conservation measures related to water and waste. These findings underscore the need to align strategic planning with actionable initiatives to foster a cohesive approach to sustainability across all stakeholder groups.

4.2.2.3 Category 3: economic viability and affordability This section examines five sustainability indicators related to economic viability and affordability. The findings, outlined in Table 4, offer a comparative analysis of stakeholder priorities. The top-ranked indicator for each group is highlighted in bold red within the table, and Fig. 5 visually depicts the variations in RII values among the three groups.

The results indicate that all stakeholders rated “Funding/Grants and contracts for sustainability development (SD) research” as the most important indicator. This is reflected in the highest RII values for this indicator: 0.771 for students, 0.733 for educators, and 0.828 for AGO staff. While students and educators rated this indicator as having high to medium importance, AGO staff assigned it high importance, emphasizing its critical role in advancing sustainability initiatives.

Table 4 further highlights that AGO staff also gave significant importance to “Financial assistance from government” (RII=0.828) and “Financial/non-financial support and incentive programs to students” (RII=0.821), underlining their focus on ensuring economic sustainability through resource allocation and support mechanisms. Educators and students, on the other hand, showed relatively balanced ratings across most indicators, with slightly lower emphasis on “Policies for affordability for low-income students”, rated 0.650 and 0.671, respectively.

Figure 5 illustrates that AGO staff consistently assigned higher RII values to all indicators compared to students and educators, likely reflecting their involvement in strategic planning and resource management. Students placed a strong emphasis on funding for SD research but assigned lower importance to broader affordability measures, such as “Policies

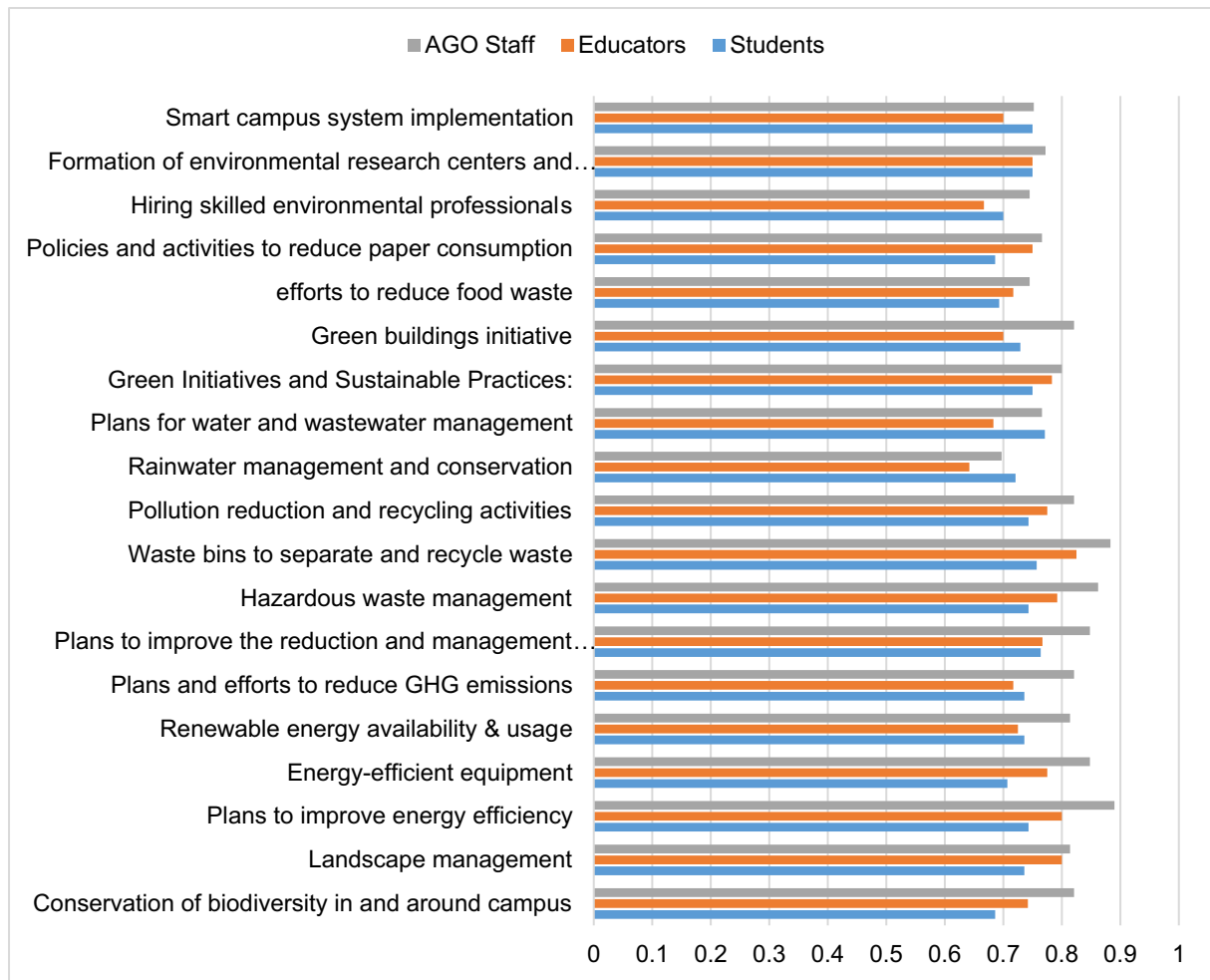


Fig. 4 Chart showing RII for category 2 sustainability indicators

Table 4 RII for various sustainability indicators by stakeholders- Category 3

No	Indicators: Category 3	Students	Educators	AGO Staff
1	Policies for affordability for low-income students	0.671	0.650	0.814
2	Financial assistance from government	0.693	0.717	0.828
3	Financial/non-financial support and incentive programs to students	0.700	0.717	0.821
4	Ethical and life cycle costing systems	0.686	0.708	0.766
5	Funding/Grants & contracts for SD research	0.771	0.733	0.828

for affordability for low-income students" (RII = 0.671). This difference highlights a potential gap in awareness or engagement among students regarding broader economic sustainability policies.

While stakeholders agree on the importance of funding and grants for SD research, the AGO staff's greater emphasis on government financial assistance and incentive programs highlights their forward-thinking approach to economic sustainability. These findings emphasize the necessity of enhancing communication and engagement efforts to align stakeholder perspectives on critical economic sustainability measures.

4.2.2.4 Category 4: academic integration Twenty sustainability indicators related to academic integration are examined in this section. The results, presented in Table 5, provide a comparative analysis of stakeholder priorities. The top-ranked

Fig. 5 Chart showing RII for category 3 sustainability indicators

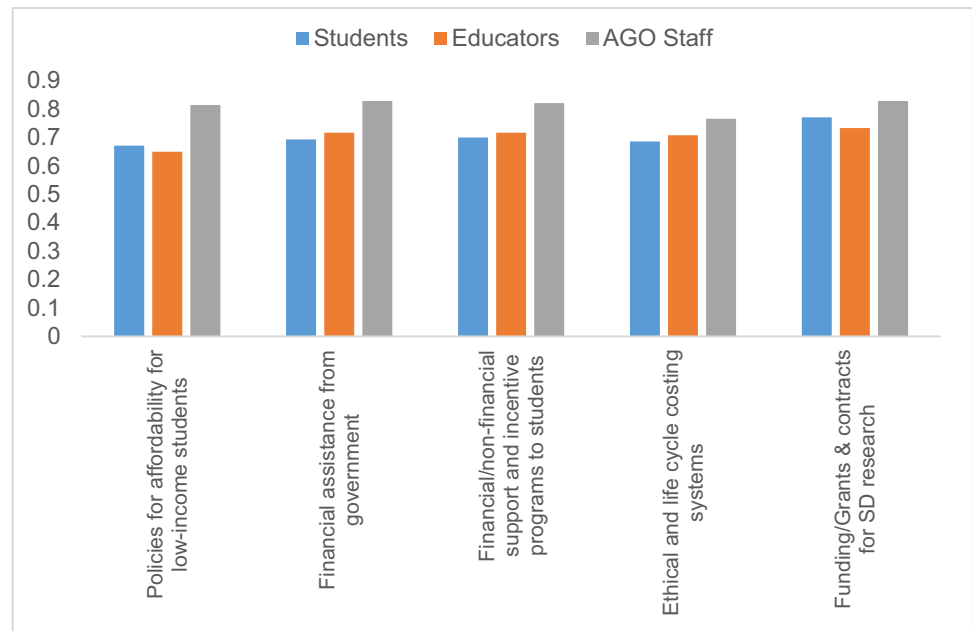


Table 5 RII for various sustainability indicators by stakeholders- Category 4

No	Indicators: Category 4	Students	Educators	AGO Staff
1	Integrated sustainability courses throughout all curricula (mandatory and optional)	0.686	0.708	0.807
2	Graduate program on sustainability	0.693	0.675	0.814
3	Availability of courses for e-learning	0.679	0.750	0.800
4	Continuing education to the public on SD	0.679	0.667	0.814
5	Courses to educate the educators on SD teaching	0.686	0.675	0.800
6	Sustainability literacy assessment for students	0.650	0.667	0.786
7	Experiential learning	0.664	0.667	0.793
8	Soft skills training activities	0.621	0.708	0.793
9	Use of research generated in SD teaching	0.650	0.700	0.800
10	Promotion of holistic thinking in research	0.657	0.675	0.800
11	Staff expertise on sustainability (attend an event or course)	0.664	0.667	0.759
12	SD institute research center	0.636	0.692	0.766
13	Publications on SD	0.650	0.775	0.821
14	Patents in the field of SD	0.636	0.700	0.779
15	Creation of SD new knowledge and technologies	0.657	0.717	0.786
16	Open access to research	0.707	0.733	0.786
17	Invited and permanent Staff dedicated to SD	0.679	0.708	0.828
18	Courses for vocational/academic training for employees	0.671	0.667	0.779
19	Teacher's performance, expertise, punctuality, and respectfulness	0.686	0.767	0.766
20	Infrastructure and Campus Environment	0.721	0.750	0.814

indicator for each group is marked in bold red within the table, while Fig. 6 visually depicts the differences in RII values across the three groups.

Significant differences in stakeholder priorities for category 4 indicators are evident from the analysis. With an RII of 0.721, students placed the highest importance on "Infrastructure and Campus Environment," underscoring their emphasis on the physical aspects of the learning environment. Educators prioritized "Publications on Sustainability Development" (RII = 0.775), reflecting their focus on academic contributions to sustainability research. Meanwhile,

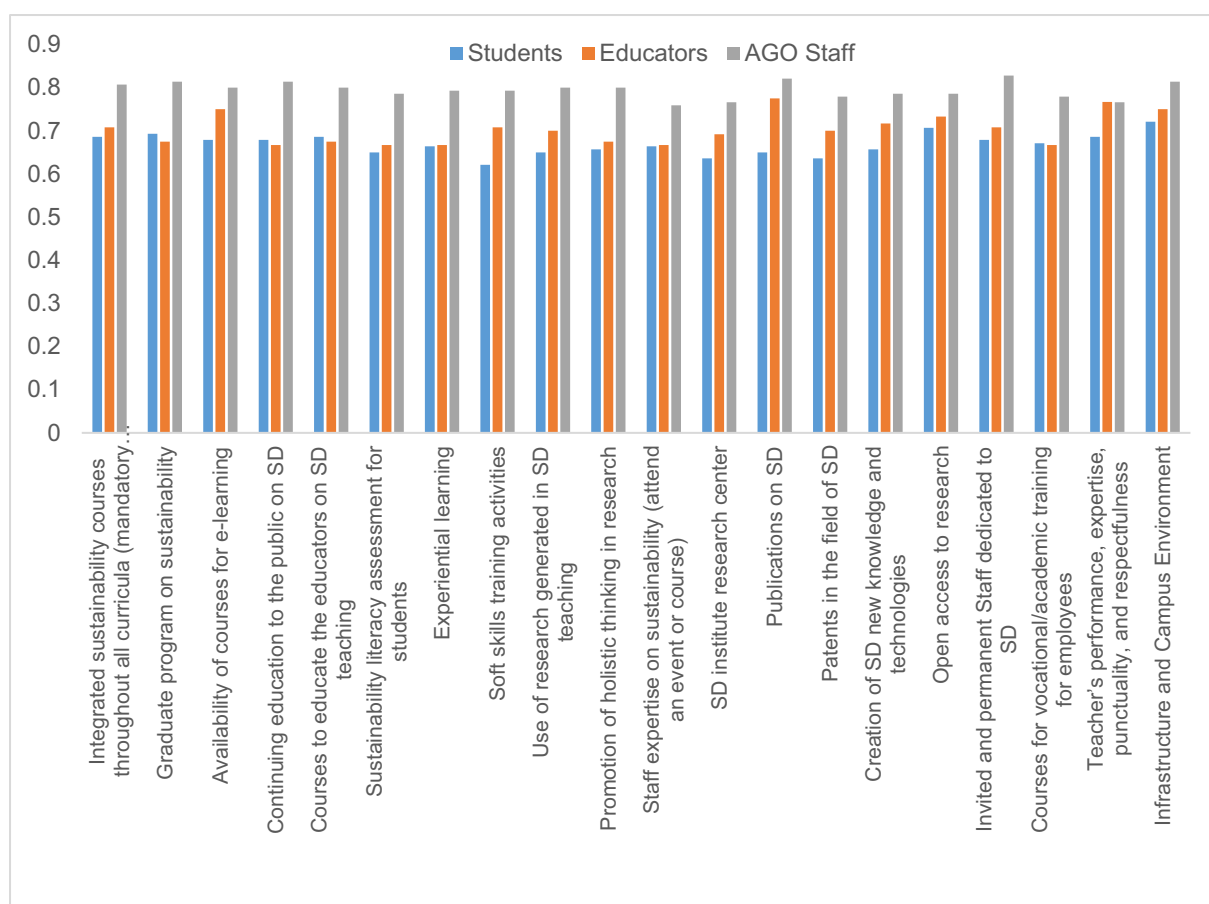


Fig. 6 Chart showing RII for category 4 sustainability indicators

AGO staff rated “Invited and permanent staff dedicated to sustainability development” as the most critical indicator (RII = 0.828), highlighting their belief in the importance of specialized staff in advancing sustainability efforts.

Figure 6 shows clear variations in the RII values across the three stakeholder groups. AGO staff consistently assigned higher RII values across most indicators, especially those related to strategic and operational aspects such as “Graduate program on sustainability” (RII = 0.814) and “Infrastructure and Campus Environment” (RII = 0.814). Educators demonstrated a balanced prioritization, giving notable emphasis to indicators like “Open access to research” (RII = 0.733) and “Teacher’s performance, expertise, punctuality, and respectfulness” (RII = 0.767), which align with their roles as academic facilitators. Students showed comparatively lower engagement with certain indicators, such as “SD institute research center” (RII = 0.636), suggesting a potential gap in their awareness or interest in institutional-level sustainability efforts.

The integration of sustainability into curricula has been widely recognized as a crucial step in fostering responsible management education. Interdisciplinary approaches play a key role in enhancing sustainability education, as they allow for a more comprehensive and applied understanding of sustainability challenges. Martins et al. [69] highlight that interdisciplinary teaching methods can bridge theoretical sustainability concepts with practical applications, ensuring students develop both critical and instrumental perspectives on sustainability. This aligns with the study’s findings, where “Integrated sustainability courses throughout all curricula” (RII = 0.807 for AGO staff) and “Graduate program on sustainability” (RII = 0.814 for AGO staff) were among the most prioritized indicators. The incorporation of interdisciplinary approaches into sustainability education frameworks can help institutions strengthen responsible management education and equip students with the necessary skills to address sustainability challenges effectively.

The AGO staff’s prioritization of dedicated sustainability staff aligns with their interview responses, which emphasized the importance of assigning responsibility to specialized personnel in each department to enhance the effectiveness of sustainability implementation. This approach is seen as a foundation for providing better services to educators and students, thereby creating a more cohesive sustainability framework.

Table 6 RII for various sustainability indicators by stakeholders- Category 5; Source: Author

No	Indicators: Category 5	Students	Educators	AGO Staff
1	Healthy lifestyle promotion and health services	0.757	0.767	0.862
2	Healthy and ergonomic work environments	0.707	0.742	0.821
3	Diversity and equity	0.793	0.750	0.834
4	Gender equality	0.793	0.708	0.883
5	Exchange programs in SD	0.779	0.617	0.821
6	Joint SD degrees with other universities	0.679	0.575	0.828
7	Joint SD research with other universities	0.736	0.617	0.834
8	SD partnerships with other society stakeholders	0.721	0.642	0.841
9	Part of interdisciplinary SD expert networks	0.664	0.608	0.828
10	Collaboration with NGOs and consultants to anticipate future needs and demands	0.771	0.650	0.834
11	Academic staff involved in voluntary advisory activities in SD	0.693	0.658	0.821
12	SD events that are open to the public/community	0.757	0.683	0.848
13	Knowledge sharing using media (email, bulletin, conference)	0.743	0.717	0.862
14	Focus on social, environmental, and economic sustainability	0.786	0.708	0.841
15	Sustainable practices for students	0.750	0.708	0.834
16	Student participation in SD activities	0.771	0.700	0.848
17	Employees and other stakeholders act sustainably	0.700	0.683	0.800
18	SD efforts are visible throughout the campus	0.721	0.733	0.821
19	Staff and researchers are aware of SD	0.764	0.717	0.834
20	Training, counseling, prevention, and risk-control programs	0.764	0.683	0.814
21	Cultural or scientific initiatives outside the community	0.721	0.667	0.772
22	SD working group with members from different departments	0.771	0.675	0.793

4.2.2.5 Category 5: social responsibility and equity This section evaluates 22 sustainability indicators under the category of social responsibility and equity. The results, summarized in Table 6, offer a comparative view of stakeholder priorities, with the highest-ranked indicator for each group marked in bold red. Figure 7 visually illustrates the variation in RII values for these indicators across the three groups.

The analysis reveals some alignment among stakeholders but also notable disparities in priorities. Students and AGO staff rated “Gender equality” as the most important indicator, with RII values of 0.793 and 0.883, respectively. Students also highlighted “Diversity and equity” as equally significant (RII = 0.793), reflecting their emphasis on inclusivity. In contrast, educators prioritized “Healthy lifestyle promotion and health services” (RII = 0.767), a perspective also strongly supported by AGO staff (RII = 0.862) and students (RII = 0.757), who assigned it their second-highest rating.

Figure 7 shows that AGO staff consistently assigned higher RII values across most indicators compared to students and educators, particularly for indicators like “Gender equality”, “Collaboration with NGOs and consultants to anticipate future needs and demands”, and “Knowledge sharing using media”. This reflects AGO staff’s operational focus on strategic partnerships, inclusivity, and knowledge dissemination. Educators demonstrated moderate prioritization across all indicators, with their highest ratings focusing on health-related and community-oriented aspects. Students, while rating some indicators highly, such as “Student participation in SD activities” (RII = 0.771) and “Focus on social, environmental, and economic sustainability” (RII = 0.786), showed relatively lower engagement with collaborative or institutional-level initiatives, such as “Part of interdisciplinary SD expert networks” (RII = 0.664).

Despite these differences, the high ratings for “Healthy lifestyle promotion and health services” across all groups underscore the shared recognition of health and well-being as foundational to sustainability. However, the disparity in ratings for other indicators highlights the need for better alignment among stakeholders to ensure a cohesive and effective approach to implementing sustainability initiatives.



Fig. 7 Chart showing RII for category 5 sustainability indicators

4.3 Discussion of responses from interviews and surveys

To effectively address the disparities in stakeholders' understanding of sustainability indicators within higher education institutions, a structured and collaborative approach is essential. This discussion integrates insights from interviews and surveys, proposing targeted strategies to enhance sustainability practices at the university.

Interviews with policymakers highlighted a lack of consistent awareness regarding sustainability rankings and indicators. Only a subset of interviewees demonstrated familiarity with these concepts, emphasizing the need for better communication and education across departments. Furthermore, while environmental sustainability initiatives were widely recognized, the social and economic dimensions received less attention, pointing to gaps in a holistic understanding. Contextual factors such as institutional structure, national policies, and socio-economic conditions play a crucial role in shaping sustainability implementation strategies. Martins et al. [70] highlight that awareness, strategic planning, and external partnerships are key drivers of sustainability integration, particularly in transitioning economies, where institutional frameworks often require localized adaptations. Aligning with these insights, this study underscores the importance of targeted awareness campaigns and structured engagement with both internal and external stakeholders to enhance sustainability implementation.

A key challenge identified in this study is resistance to technological change, which manifests in several ways within HEIs. First, a lack of digital infrastructure often hinders the seamless adoption of sustainable technologies, particularly in institutions with constrained budgets or outdated systems. Second, digital literacy gaps among faculty, staff, and students can slow down the adoption of sustainability-focused technological solutions, as some stakeholders may lack the necessary training or familiarity with emerging digital tools. Third, cultural resistance to change presents another significant barrier, where traditional institutional practices and long-standing operational procedures can create inertia, making faculty and staff hesitant to transition to more technology-driven sustainability initiatives. Overcoming these barriers requires a multifaceted approach, including targeted capacity-building programs to enhance digital literacy, institutional incentives to encourage the use of sustainable technologies, and a phased implementation strategy that allows for gradual adaptation. Additionally, fostering a culture of technological openness through leadership advocacy and cross-departmental collaboration can help reduce resistance and promote a more inclusive and effective integration of technology into sustainability efforts.

Surveys revealed significant variations in the prioritization of sustainability indicators among students, educators, and AGO staff. For instance, AGO staff prioritized governance and strategic planning, while students and educators focused on tangible outcomes like improved infrastructure and health services. Such differences highlight a divergence in roles, responsibilities, and awareness levels across stakeholder groups. Innovative pedagogical methods, such as experiential and interdisciplinary learning approaches, can help bridge these gaps by fostering a more engaged and proactive sustainability culture. Meyer et al. [71] suggest that structured innovation communities and open learning environments, such as the “Sandbox Approach,” enable students to collaboratively develop sustainable solutions while addressing real-world challenges. Incorporating similar approaches within the university’s sustainability education framework could enhance stakeholder engagement and promote a more integrated perspective on sustainability challenges.

To enhance sustainability initiatives and their implementation at the university, several key recommendations are proposed. Stakeholder engagement and collaboration are critical, with regular interdisciplinary workshops and cross-functional working groups ensuring diverse perspectives are integrated into strategies. Communication and education should focus on developing accessible resources that explain sustainability indicators and providing continuous updates through newsletters and dashboards to keep stakeholders informed.

Customizing sustainability frameworks to align with the university’s unique culture and priorities is essential, with participatory approaches like surveys and focus groups fostering stakeholder input and validation. Capacity building, through embedding sustainability topics in curricula and professional development programs, and peer mentorship programs can further promote shared learning and a culture of sustainability.

Technology can enhance engagement and accessibility by utilizing interactive dashboards and online knowledge repositories for sustainability data and resources. Continuous monitoring, regular assessments, and benchmarking against similar institutions will support ongoing improvements and highlight best practices.

Leadership and cultural integration are fundamental, with university leadership championing sustainability as a core value and embedding it into institutional identity, operations, and decision-making.

While the recommendations outlined in this study provide a structured approach to enhancing sustainability implementation in HEIs, the practical execution of these strategies presents significant challenges. Financial constraints remain a key barrier, as many institutions struggle to allocate dedicated budgets for sustainability efforts amid competing institutional priorities. To overcome this challenge, HEIs can explore alternative funding sources, such as government grants, industry partnerships, and philanthropic contributions, to support sustainability projects. Additionally, phased implementation, prioritizing cost-effective initiatives with high impact, such as digitalization and energy-efficient practices, can facilitate gradual progress while minimizing financial strain. Resistance to technological change is another critical hurdle, often stemming from a lack of digital literacy, skepticism regarding new systems, or institutional inertia. Addressing this requires a multi-pronged approach, including targeted training programs to enhance stakeholders’ confidence in digital tools, active engagement of faculty and staff in decision-making to foster a sense of ownership, and strong leadership advocacy to emphasize the long-term benefits of technological integration. By implementing these strategies, HEIs can create a more resilient and adaptable framework for sustainability, ensuring the successful execution of interdisciplinary working groups, centralized reporting platforms, and long-term sustainability governance.

5 Conclusion

This research emphasizes the critical role of HEIs in advancing sustainability by utilizing structured frameworks of indicators and fostering stakeholder engagement. Through an extensive literature review, a comprehensive set of sustainability indicators was identified, encompassing environmental, social, economic, governance, and academic pillars. These indicators provide a robust foundation for assessing sustainability in HEIs but also reveal disparities in stakeholder perceptions, highlighting the need for improved communication, alignment, and transparency.

Insights from semi-structured interviews with policymakers uncovered key challenges that hinder effective sustainability implementation. These included cost-sustainability trade-offs, resistance to technological change, and fragmented coordination in sustainability reporting. Financial constraints often forced HEIs to balance sustainability initiatives against competing institutional priorities, creating a delicate challenge for effective implementation. These findings align with existing literature, further underscoring the systemic barriers to embedding sustainability in institutional operations.

The survey-based RII analysis shed light on the differing priorities among stakeholders. Administrative staff primarily emphasized strategic governance and long-term planning, while students and educators focused on tangible outcomes such as improved infrastructure and health services. These differences reflect the unique roles and responsibilities of each group but also reveal the need for a unified vision to align diverse stakeholder perspectives.

The divergence in priorities and perspectives underscores the complexities of embedding sustainability within HEIs. A recurring theme of this research is the importance of adopting a holistic and participatory approach to sustainability. Bridging the gap between strategic governance and practical implementation requires enhanced collaboration among stakeholders and the integration of diverse perspectives into actionable strategies that resonate across all groups.

The study also highlights the necessity of moving beyond a purely environmental focus to address interconnected pillars of governance, social equity, and economic viability. A balanced approach is crucial for fostering inclusive, transparent, and effective sustainability frameworks. Additionally, regular evaluations and tailored frameworks are vital for ensuring that sustainability strategies remain relevant and adaptive to changing institutional and global contexts.

This study offers valuable insights into sustainability in HEIs; however, certain limitations must be acknowledged. While the sample size for the RII analysis was sufficient for statistical assessment, expanding it would enhance the generalizability of the findings and provide a more comprehensive understanding of stakeholder perspectives across different HEIs. Additionally, although the RII analysis effectively highlighted meaningful variations and consistent trends in stakeholder priorities, statistical significance tests were not conducted to confirm these differences. Future research should incorporate parametric or non-parametric tests to enhance the robustness of the findings. Nonetheless, the current results remain valuable, as the RII findings align with the qualitative insights from interviews, reinforcing the patterns observed. This convergence between quantitative and qualitative data supports the internal consistency and relevance of the study's outcomes. Moreover, the study's reliance on convenience sampling in survey distribution may have introduced selection bias, potentially affecting the representativeness of stakeholder viewpoints. To enhance the reliability of future studies, employing randomized sampling techniques would improve the accuracy and applicability of results across a broader institutional landscape.

Another key limitation is the study's reliance on a single case study. While the findings offer valuable insights, their applicability to HEIs in different geographic, cultural, and institutional contexts may be limited. The unique governance structures, policies, and sustainability priorities of other institutions may lead to different outcomes. Future research should explore multi-institutional studies across diverse regions to enhance the generalizability of sustainability assessment frameworks and identify best practices that can be adapted to various HEI settings.

Despite these limitations, this research serves as a roadmap for HEIs to advance their sustainability initiatives, aligning their efforts with the broader global agenda of the SDGs. By fostering stakeholder collaboration, improving sustainability literacy, and aligning institutional strategies with the priorities identified in this study, HEIs can transition from fragmented efforts to cohesive and impactful approaches. Future research should build upon these findings by exploring longitudinal studies, technological advancements in sustainability tracking, and comparative assessments across multiple HEIs to further refine sustainability frameworks and practices.

To enhance the implementation of sustainability initiatives in HEIs, the following recommendations are proposed:

- Facilitate regular and structured dialogue among students, educators, and AGO staff to align priorities and ensure collaborative decision-making. Establish interdisciplinary working groups to foster inclusivity and shared ownership of sustainability goals.

- Develop targeted training programs, workshops, and seminars to improve sustainability literacy across all stakeholder groups. These initiatives should focus on the practical application of sustainability principles and their relevance to institutional objectives.
- Create centralized platforms, such as interactive dashboards and online portals, for sustainability reporting and progress tracking. These tools should provide stakeholders with real-time insights into achievements, challenges, and upcoming initiatives, promoting transparency and accountability.
- Adapt global sustainability frameworks to reflect the unique needs, priorities, and cultural contexts of individual HEIs. Involve stakeholders in this customization process to ensure the relevance and practicality of the frameworks.
- Employ advanced digital tools and data analytics to streamline sustainability assessments, automate reporting processes, and monitor progress in real time. Visualization tools can also aid in communicating complex data to a wider audience effectively.
- Establish mechanisms for regular assessment of sustainability strategies, incorporating feedback from stakeholders to refine approaches. This iterative process ensures adaptability to evolving challenges, opportunities, and institutional goals.
- Promote partnerships between academic disciplines to integrate sustainability principles into teaching, research, and community engagement. Such collaboration encourages innovation and supports the development of holistic solutions to sustainability challenges.

By implementing these recommendations, HEIs can create a cohesive and inclusive framework for sustainability, enhancing their capacity to contribute meaningfully to global sustainability efforts while addressing their unique institutional needs.

Building on the findings and contributions of this study, future research should focus on the following areas to advance the field of sustainability in HEIs:

- Create comprehensive sustainability assessment frameworks that account for regional, institutional, and cultural variations while maintaining alignment with global standards.
- Investigate the application of cutting-edge technologies, such as the Internet of Things and Artificial Intelligence, to improve the accuracy, efficiency, and real-time monitoring of sustainability initiatives.
- Conduct longitudinal studies to evaluate the long-term effects of sustainability initiatives on institutional performance, stakeholder engagement, and satisfaction. Sustainability priorities evolve over time due to institutional changes, policy shifts, and emerging global sustainability challenges. A longitudinal approach would provide a more dynamic understanding of how HEIs adapt to these changes, how stakeholder perceptions shift, and how sustainability initiatives impact governance, academic curricula, and campus operations in the long run.
- Expand research to include cross-regional and cross-cultural comparisons of HEIs, identifying best practices and transferable strategies that address diverse sustainability challenges.
- Examine the role of HEIs in promoting sustainability beyond their institutional boundaries, shaping societal behaviors, policies, and industry practices to drive broader systemic change.
- Enhance the integration of social and economic dimensions into sustainability frameworks to ensure a balanced approach beyond environmental sustainability. Future studies should explore the inclusion of equity, diversity, and financial accessibility indicators and assess how HEIs can align social and economic sustainability with institutional strategies. Comparative studies across institutions with diverse socio-economic contexts can provide further insights into developing more holistic sustainability frameworks.

By addressing these areas, future research can deepen our understanding of sustainability in higher education, refine implementation strategies, and strengthen the position of HEIs as pivotal contributors to global sustainability efforts.

Name of the approval committee or the internal review board (IRB) Adrian Lopes, IRB Chair, American University of Sharjah.

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Declarations

Ethics approval and consent to participate The study was approved by the Institutional Review Board (IRB) of the American University of Sharjah (Protocol #: 24-072, approved on April 15, 2024). For the interviews, the ethical statement was included as part of the interview questions, and the consent of the interviewees was obtained accordingly by sending the interview questions in advance and requesting their agreement to participate through a consent form. For the survey, consent was obtained via a Google Form before accessing the survey page.

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